Problem **1**(a):

Suppose the γ^μ obey the anticommutation relations (1). Then:

$$\begin{aligned}
\beta^2 &= \gamma^0\gamma^0 = g^{00} = +1, && \text{(S.1)}\\
\{\beta,\alpha^i\} &= \{\gamma^0,\gamma^0\gamma^i\} = \gamma^0\gamma^0\gamma^i + \gamma^0\gamma^i\gamma^0 \\
&= \gamma^0\times\{\gamma^0,\gamma^i\} = \gamma^0\times 0 = 0, && \text{(S.2)}\\
\alpha^i\alpha^j &= \gamma^0\gamma^i\gamma^0\gamma^j = -\gamma^0\gamma^0\gamma^i\gamma^j = -\gamma^i\gamma^j, && \text{(S.3)}\\
&\Downarrow \\
\{\alpha^i,\alpha^j\} &= -\{\gamma^i,\gamma^j\} = -2g^{ij} = +2\delta^{ij}, && \text{(S.4)}
\end{aligned}$$

quod erat demonstrandum.

Conversely, suppose $\beta^2 = 1$, $\{\beta,\alpha^i\} = 0$, and $\{\alpha^i,\alpha^j\} = +2\delta^{ij}$, and let us *define* $\gamma^0 \stackrel{\text{def}}{=} \beta$ and $\gamma^i \stackrel{\text{def}}{=} \beta\alpha^i$. In this case:

$$\begin{aligned}
\gamma^0\gamma^0 &= \beta\beta = +1 = g^{00}, && \text{(S.5)}\\
\{\gamma^0,\gamma^i\}\ \{\beta,\beta\alpha^i\} &= \beta\times\{\beta,\alpha^i\} = \beta\times 0 = 0 = 2g^{0i}, && \text{(S.6)}\\
\gamma^i\gamma^j &= \beta\alpha^i\beta\alpha^j = -\beta\beta\alpha^i\alpha^j = -\alpha^i\alpha^j, && \text{(S.7)}\\
&\Downarrow \\
\{\gamma^i,\gamma^j\} &= -\{\alpha^i,\alpha^j\} = -2\delta^{ij} = +2g^{ij}, && \text{(S.8)}
\end{aligned}$$

and therefore

$$\forall\mu,\nu:\quad \{\gamma^\mu,\gamma^\nu\} = 2g^{\mu\nu}\,. \qquad (1)$$

Quod erat demonstrandum.

Problem $\mathbf{1}$(b):

$$\gamma^\alpha\gamma_\alpha = \tfrac{1}{2}\{\gamma^\alpha,\gamma^\beta\}g_{\alpha\beta} = g^{\alpha\beta}g_{\alpha\beta} = 4; \tag{S.9}$$

$$\begin{aligned}\gamma^\alpha\gamma^\nu\gamma_\alpha &= (\gamma^\alpha\gamma^\nu = 2g^{\nu\alpha} - \gamma^\nu\gamma^\alpha)\gamma_\alpha\\ &= 2\gamma^\nu - \gamma^\nu(\gamma^\alpha\gamma_\alpha = 4) = -2\gamma^\nu;\end{aligned} \tag{S.10}$$

$$\begin{aligned}\gamma^\alpha\gamma^\mu\gamma^\nu\gamma_\alpha &= (\gamma^\alpha\gamma^\mu = 2g^{\mu\alpha} - \gamma^\mu\gamma^\alpha)\gamma^\nu\gamma_\alpha\\ &= 2\gamma^\nu\gamma^\mu - \gamma^\mu(\gamma^\alpha\gamma^\nu\gamma_\alpha = -2\gamma^\nu)\\ &= 2\gamma^\nu\gamma^\mu + 2\gamma^\mu\gamma^\nu = 4g^{\mu\nu};\end{aligned} \tag{S.11}$$

$$\begin{aligned}\gamma^\alpha\gamma^\lambda\gamma^\mu\gamma^\nu\gamma_\alpha &= (\gamma^\alpha\gamma^\lambda = 2g^{\mu\alpha} - \gamma^\lambda\gamma^\alpha)\gamma^\mu\gamma^\nu\gamma_\alpha\\ &= 2\gamma^\mu\gamma^\nu\gamma^\lambda - \gamma^\lambda(\gamma^\alpha\gamma^\mu\gamma^\nu\gamma_\alpha = 4g^{\mu\nu})\\ &= (2\gamma^\mu\gamma^\nu - 4g^{\mu\nu} = -2\gamma^\nu\gamma^\mu)\gamma^\lambda\\ &= -2\gamma^\nu\gamma^\mu\gamma^\lambda.\end{aligned} \tag{S.12}$$

Problem $\mathbf{1}$(c):

First, a Lemma: for any Lorentz vector a^μ, the $\not{a} \stackrel{\text{def}}{=} \gamma^\mu a_\mu$ matrix squares to

$$\not{a}\,\not{a} = \gamma^\mu a_\mu\gamma^\nu a_\nu = a_\mu a_\nu \times \left(\gamma^\mu\gamma^\nu = g^{\mu\nu} - 2iS^{\mu\nu}\right) = a^2 - i[a_\mu, a_\nu]\times S^{\mu\nu} \tag{S.13}$$

(where the last equality comes from $S^{\mu\nu} = -S^{\nu\mu}$.) For vectors a^μ whose components commute with each other, this formula simplifies to $\not{a}^2 = a^2$, in particular for the ordinary derivatives ∂^μ, $\not{\partial}^2 = \partial^2$. However, the covariant derivatives D_μ do not commute with each other. Instead, $[D_\mu, D_\nu] = iqF_{\mu\nu}(x)$ where q is the electric charge of the field on which D_μ act; for the electron field $\Psi(x)$, $q = -e$ and hence $[D_\mu, D_\nu]\Psi(x) = -ieF_{\mu\nu}(x)\Psi(x)$. Hence, according to the lemma (S.13),

$$\not{D}^2\Psi = D^2\Psi - eF_{\mu\nu}S^{\mu\nu}\Psi. \tag{S.14}$$

Now, suppose the electron field $\Psi(x)$ satisfies the covariant Dirac equation $(i\not{D} - m)\Psi = 0$.

Then for any differential operator $\mathcal{D}$, $\mathcal{D} \times (i\not{D} - m)\Psi = 0$, and in particular

$$(-i\not{D} - m) \times (i\not{D} - m)\Psi \;=\; 0. \tag{S.15}$$

The LHS of this formula amounts to

$$(-i\not{D} - m) \times (i\not{D} - m)\Psi \;=\; \big(\not{D}^2 + m^2\big)\Psi \;=\; \big(D^2 - eF_{\mu\nu}S^{\mu\nu} + m^2\big)\Psi, \tag{S.16}$$

which immediately leads to eq. (2).

Problem **1**(d):

The anti-commutation relations (1) imply $\gamma^\mu\gamma^\nu = \pm\gamma^\nu\gamma^\mu$ where the sign is '+' for $\mu = \nu$ and '$-$' otherwise. Hence for any product Γ of the γ matrices, $\gamma^\mu\Gamma = (-1)^n\Gamma\gamma^\mu$, where n is the number of $\gamma^{\nu\neq\mu}$ factors of Γ. For the $\Gamma = \gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3$, $n = 3$ for any $\mu = 0, 1, 2, 3$, hence $\gamma^\mu\gamma^5 = -\gamma^5\gamma^\mu$.

As to the spin matrices, $\gamma^5\gamma^\mu\gamma^\nu = -\gamma^\mu\gamma^5\gamma^\nu = +\gamma^\mu\gamma^\nu\gamma^5$ and therefore $\gamma^5 S^{\mu\nu} = +S^{\mu\nu}\gamma^5$.

Problem **1**(e):

First, the hermiticity:

$$\begin{aligned}
\big(\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3\big)^\dagger \;&=\; -i(\gamma^3)^\dagger(\gamma^2)^\dagger(\gamma^1)^\dagger(\gamma^0)^\dagger \;=\; -i(-\gamma^3)(-\gamma^2)(-\gamma^1)(+\gamma^0) \\
&=\; +i\gamma^3\gamma^2\gamma^1\gamma^0 \;=\; +i(\gamma^3\gamma^2\gamma^1)\times\gamma^0 \;=\; +i(-1)^3\gamma^0\times\gamma^3\gamma^2\gamma^1 \\
&=\; +i(-1)^3\gamma^0\times(-1)^2\gamma^1\times\gamma^3\gamma^2 \;=\; +i(-1)^3\gamma^0\times(-1)^2\gamma^1\times(-1)\gamma^2\gamma^1 \\
&=\; (-1)^6\times(+i\gamma^0\gamma^1\gamma^2\gamma^3) \;\equiv\; +1\times\gamma^5.
\end{aligned} \tag{S.17}$$

Second, the square:

$$\begin{aligned}
(\gamma^5)^2 \;&=\; \gamma^5(\gamma^5)^\dagger \;=\; (i\gamma^0\gamma^1\gamma^2\gamma^3)(i\gamma^3\gamma^2\gamma^1\gamma^0) \;=\; -\gamma^0\gamma^1\gamma^2(\gamma^3\gamma^3)\gamma^2\gamma^1\gamma^0 \\
&=\; +\gamma^0\gamma^1(\gamma^2\gamma^2)\gamma^1\gamma^0 = \; -\gamma^0(\gamma^1\gamma^1)\gamma^0 \;=\; +\gamma^0\gamma^0 \;=\; +1.
\end{aligned} \tag{S.18}$$

Problem $\mathbf{1}$(f):

Since the four Dirac matrices γ^0, γ^1, γ^2, γ^3 all anticommute with each other, we have

$$\epsilon_{\kappa\lambda\mu\nu}\gamma^\kappa\gamma^\lambda\gamma^\mu\gamma^\nu \;=\; \gamma^{[0}\gamma^1\gamma^2\gamma^{3]} \;=\; 24\gamma^0\gamma^1\gamma^2\gamma^3 \;=\; -24i\gamma^5. \tag{S.19}$$

To prove the other identity, we note that a totally antisymmetric product $\gamma^{[\kappa}\gamma^\lambda\gamma^\mu\gamma^{\nu]}$ vanishes unless the Lorentz indices $\kappa, \lambda, \mu, \nu$ are all distinct — which makes them $0, 1, 2, 3$ in some order. For such indices, the anticommutativity of the Dirac matrices implies $\gamma^\kappa\gamma^\lambda\gamma^\mu\gamma^\nu = -\epsilon^{\kappa\lambda\mu\nu} \times \gamma^0\gamma^1\gamma^2\gamma^3$ (note that $\epsilon^{0123} = -1$), and hence

$$\gamma^{[\kappa}\gamma^\lambda\gamma^\mu\gamma^{\nu]} \;=\; -24\epsilon^{\kappa\lambda\mu\nu} \times \gamma^0\gamma^1\gamma^2\gamma^3 \;=\; +24i\epsilon^{\kappa\lambda\mu\nu} \times \gamma^5. \tag{S.20}$$

Problem $\mathbf{1}$(g):

$$\begin{aligned}
6i\epsilon^{\kappa\lambda\mu\nu}\,\gamma_\kappa\gamma^5 &= \tfrac{6}{24}\,\gamma_\kappa\,\gamma^{[\kappa}\gamma^\lambda\gamma^\mu\gamma^{\nu]} \\
&= \tfrac{1}{4}\,\gamma_\kappa\,\Big(\gamma^\kappa\gamma^{[\lambda}\gamma^\mu\gamma^{\nu]} \;-\; \gamma^{[\lambda)}\gamma^\kappa\gamma^{(\mu}\gamma^{\nu]} \;+\; \gamma^{[\lambda}\gamma^{\mu)}\gamma^\kappa\gamma^{(\nu]} \;-\; \gamma^{[\lambda}\gamma^\mu\gamma^{\nu]}\gamma^\kappa\Big) \\
&= \tfrac{1}{4}\,\Big(4\gamma^{[\lambda}\gamma^\mu\gamma^{\nu]} \;+\; 2\gamma^{[\lambda}\gamma^\mu\gamma^{\nu]} \;+\; 4g^{[\lambda\mu}\gamma^{\nu]} \;+\; 2\gamma^{[\nu}\gamma^\mu\gamma^{\lambda]}\Big) \\
&= \tfrac{1}{4}(4+2+0-2)\times\gamma^{[\lambda}\gamma^\mu\gamma^{\nu]} \;=\; \gamma^{[\lambda}\gamma^\mu\gamma^{\nu]}.
\end{aligned} \tag{S.21}$$

Problem $\mathbf{1}$(h):

Proof by inspection: In the Weyl basis and in the 2×2 block form, the 16 matrices are

$$\begin{gathered}
\mathbf{1}_{4\times4} \;=\; \begin{pmatrix} \mathbf{1} & 0 \\ 0 & \mathbf{1} \end{pmatrix}, \qquad \gamma^0 \;=\; \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \gamma^i \;=\; \begin{pmatrix} 0 & +\sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \\
\tfrac{1}{2}\gamma^{[i}\gamma^{j]} \;=\; -i\epsilon^{ijk}\begin{pmatrix} \sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix}, \qquad \tfrac{1}{2}\gamma^{[0}\gamma^{i]} \;=\; \begin{pmatrix} -\sigma^i & 0 \\ 0 & +\sigma^i \end{pmatrix}, \\
\gamma^5\gamma^0 \;=\; \begin{pmatrix} 0 & -\mathbf{1} \\ +\mathbf{1} & 0 \end{pmatrix}, \qquad \gamma^5\gamma^i \;=\; \begin{pmatrix} 0 & -\sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \qquad \gamma^5 \;=\; \begin{pmatrix} -\mathbf{1} & 0 \\ 0 & +\mathbf{1} \end{pmatrix},
\end{gathered} \tag{S.22}$$

and their linear independence is self-evident. Since there are only 16 independent 4×4 matrices altogether, any such matrix Γ is a linear combination of the matrices (S.22). $\mathcal{Q.E.D.}$

Algebraic Proof: Without making any assumption about the matrix form of the γ^μ operators, let us consider the Clifford algebra (1). Using $\gamma^\mu\gamma^\nu = \pm\gamma^\nu\gamma^\mu$ where the sign is $+$ for $\mu = \nu$ and $-$ for $\mu \neq \nu$, we may re-order any product of the γ matrices as $\pm\gamma^0\cdots\gamma^0\,\gamma^1\cdots\gamma^1\,\gamma^2\cdots\gamma^2\,\gamma^3\cdots\gamma^3$. Moreover, since each γ^μ squares to $+1$ or -1, we may further simplify the product in question to to $\pm(\gamma^0 \text{ or } 1)\times(\gamma^1 \text{ or } 1)\times(\gamma^2 \text{ or } 1)\times(\gamma^3 \text{ or } 1)$. The net result is (up to a sign or $\pm i$ factor) one of the 16 matrices: 1, or a γ^μ, or a $\gamma^\mu\gamma^\nu$ for $\mu \neq \nu$ which equals to $\frac{1}{2}\gamma^{[\mu}\gamma^{\nu]}$, or a $\gamma^\lambda\gamma^\mu\gamma^\nu$ for 3 different λ, μ, ν which equals to $\frac{1}{6}\gamma^{[\lambda}\gamma^\mu\gamma^{\nu]} = i\epsilon^{\lambda\mu\nu\rho}\gamma^5\gamma_\rho$ (*cf.* part (g)), or $\gamma^0\gamma^1\gamma^2\gamma^3 = -i\gamma^5$. Consequently, any operator Γ algebraically constructed of the γ's is a linear combination of these 16 matrices.

Incidentally, this proof explains why the Dirac matrices are 4×4 in $d = 4$ spacetime dimensions: the 16 linearly-independent products of Dirac matrices require matrix size to be $\sqrt{16} = 4$.

Technically, we may also use matrices of size $4n \times 4n$, but then we would have $\gamma^\mu = \gamma^\mu_{4\times 4} \otimes \mathbf{1}_{n\times n}$, and ditto for all their products. Physically, this means combining the Dirac spinor index with some other index $i = 1, \ldots, n$ which has nothing to do with Lorentz symmetry. Nobody wants such an index confusion, so physicists always stick to 4×4 Dirac matrices in 4 spacetime dimensions.

Problem **2**:

As in problem 1(d), for any product Γ of the Dirac matrices, $\gamma^\mu\Gamma = (-1)^n\Gamma\gamma^\mu$ where n is the number of $\gamma^{\nu\neq\mu}$ factors of Γ. For $\Gamma = \gamma^0\gamma^1\cdots\gamma^{d-1}$ (modulo an overall sign or a $\pm i$ factor), each γ^μ appear once, with the remaining $d-1$ factors being $\gamma^{\nu\neq\mu}$. Thus, $n = d-1$ and $\gamma^\mu\Gamma = (-1)^{d-1}\Gamma\gamma^\mu$: for even spacetime dimensions d, Γ anticommutes with all the γ^μ matrices, but for odd d, Γ commutes with all the γ^μ instead of anticommuting.

Now consider the dimension of the Clifford algebra made out of Dirac matrix products and their linear combinations. As in problem 1(h), any product of Dirac matrices can be re-arranged as $\pm(\gamma^0 \text{ or } 1)\times(\gamma^1 \text{ or } 1)\times\cdots\times(\gamma^{d-1} \text{ or } 1)$, and the number of distinct products of this type is clearly 2^d. Alternatively, we count a single unit matrix, $\binom{d}{1} = d$ of γ^μ matrices, $\binom{d}{2}$ of independent $\frac{1}{2}\gamma^{[\mu}\gamma^{\nu]}$ products, $\binom{d}{3}$ of $\frac{1}{6}\gamma^{[\lambda}\gamma^\mu\gamma^{\nu]}$ products, *etc.*, *etc.*, all the way up to

$\binom{d}{d-1} = d$ of $\gamma^{[\beta}\cdots\gamma^{\omega]} \propto \epsilon^{\alpha\beta\cdots\omega}\gamma_\alpha\Gamma$, and $\binom{d}{d} = 1$ of $\gamma^{[\alpha}\cdots\gamma^{\omega]} \propto \epsilon^{\alpha\beta\cdots\omega}\Gamma$; the net number of all such matrices is $\sum_{k=0}^{d}\binom{d}{k} = 2^d$.

For even dimensions d, all the 2^d matrix products are distinct, and we need matrices of size $2^{d/2} \times 2^{d/2}$ to accommodate them. But for odd d, the Γ matrix commutes with the entire Clifford algebra (since it commutes with all the γ^μ), so to avoid redundancy we should set $\Gamma = 1$ or $\Gamma = -1$. Consequently, any product of k distinct Dirac matrices becomes identical (up to a sign or a $\pm i$) with the product of the other $d - k$ matrices, for example $\gamma^0 \cdots \gamma^{k-1} = i^{??}\gamma^k \cdots \gamma^{d-1}$. This halves the net dimension of the Clifford algebra from 2^d to 2^{d-1} and calls for matrix size $2^{(d-1)/2} \times 2^{(d-1)/2}$.

Thus, the Dirac matrices in 2 or 3 space time dimensions are 2×2, in 4 or 5 dimensions they are 4×4, in 6 or 7 dimensions — 8×8, in 8 or 9 dimensions — 16×16, in 10 or 11 dimensions — 32×32, *etc.*, *etc.*

Problem **3**(a):

I have already written down the γ^5 and the $\frac{1}{2}\gamma^{[\mu}\gamma^{\nu]} = -2iS^{\mu\nu}$ in the Weyl basis. (S.22), but here is the detailed calculation, in case you need it:

$$\begin{aligned}
\gamma^5 &= i\begin{pmatrix} 0 & \sigma^0 \\ \overline{\sigma}^0 & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma^1 \\ \overline{\sigma}^1 & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma^2 \\ \overline{\sigma}^2 & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma^3 \\ \overline{\sigma}^3 & 0 \end{pmatrix} \\
&= i\begin{pmatrix} \sigma^0\overline{\sigma}^1\sigma^2\overline{\sigma}^3 & 0 \\ 0 & \overline{\sigma}^0\sigma^1\overline{\sigma}^2\sigma^3 \end{pmatrix} = \begin{pmatrix} +i\sigma^1\sigma^2\sigma^3 & 0 \\ 0 & -i\sigma^1\sigma^2\sigma^3 \end{pmatrix} \\
&= \begin{pmatrix} -1 & 0 \\ 0 & +1 \end{pmatrix}.
\end{aligned} \tag{4}$$

Problem **3**(b):

In the Weyl basis for the γ matrices,

$$\gamma^\mu\gamma^\nu = \begin{pmatrix} 0 & \sigma^\mu \\ \overline{\sigma}^\mu & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma^\nu \\ \overline{\sigma}^\nu & 0 \end{pmatrix} = \begin{pmatrix} \sigma^\mu\overline{\sigma}^\nu & 0 \\ 0 & \overline{\sigma}^\mu\sigma^\nu \end{pmatrix}, \tag{S.23}$$

and hence

$$S^{ij} \;=\; \frac{i}{4}\begin{pmatrix} -\sigma^{[i}\sigma^{j]} & 0 \\ 0 & -\sigma^{[i}\sigma^{j]} \end{pmatrix} \;=\; \frac{\epsilon^{ijk}}{2}\begin{pmatrix} \sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix} \tag{S.24}$$

while

$$S^{0k} \;=\; -S^{k0} \;=\; \tfrac{i}{2}\gamma^0\gamma^k \;=\; \frac{i}{2}\begin{pmatrix} -\sigma^k & 0 \\ 0 & +\sigma^k \end{pmatrix}. \tag{S.25}$$

By inspection, these spin matrices agree with the $\mathbf{2}+\overline{\mathbf{2}}$ reducible representation of the Lorentz generators,

$$S^{ij} \;=\; \epsilon^{ijk}\begin{pmatrix} \hat{J}^k_{\mathbf{2}} & 0 \\ 0 & \hat{J}^k_{\overline{\mathbf{2}}} \end{pmatrix} \quad \text{while} \quad S^{0i} \;=\; -S^{i0} \;=\; \begin{pmatrix} \hat{K}^i_{\mathbf{2}} & 0 \\ 0 & \hat{K}^i_{\overline{\mathbf{2}}} \end{pmatrix}, \tag{S.26}$$

or in one formula

$$S^{\mu\nu} \;=\; \begin{pmatrix} \hat{J}^{\mu\nu}_{\mathbf{2}} & 0 \\ 0 & \hat{J}^{\mu\nu}_{\overline{\mathbf{2}}} \end{pmatrix}. \tag{5}$$

Problem **3**(c):

Consider the squares of the boost representations (8),

$$\big(M_L(B)\big)^2 \;=\; \exp(-r\,\mathbf{n}\cdot\boldsymbol{\sigma}) \quad\text{and}\quad \big(M_R(B)\big)^2 \;=\; \exp(+r\,\mathbf{n}\cdot\boldsymbol{\sigma}), \tag{S.27}$$

and let's expand these matrix exponentials in powers of the rapidity r:

$$\exp(\mp r\,\mathbf{n}\cdot\boldsymbol{\sigma}) \;=\; \sum_{k=0}^{\infty}\frac{(\mp r)^k}{k!}\times(\mathbf{n}\cdot\boldsymbol{\sigma})^k. \tag{S.28}$$

But thanks to $\{\sigma^i,\sigma^j\}=2\delta^{ij}$, for any unit vector $\mathbf{n}$ we have

$$(\mathbf{n}\cdot\boldsymbol{\sigma})^2 \;=\; n^in^j\sigma^i\sigma^j \;=\; \tfrac{1}{2}n^in^j\{\sigma^i,\sigma^j\} \;=\; n^in^j\times\delta^{ij} \;=\; \mathbf{n}^2 \;=\; 1, \tag{S.29}$$

hence for any integer power $k=0,1,2,3,4,\ldots$

$$(\mathbf{n}\cdot\boldsymbol{\sigma})^k \;=\; \begin{cases} 1 & \text{for even } k, \\ (\mathbf{n}\cdot\boldsymbol{\sigma}) & \text{for odd } k. \end{cases} \tag{S.30}$$

Plugging this formula into eqs. (S.27) for the matrix exponentials and their expansions in powers of the rapidity r, we get

$$\begin{aligned}
\exp\bigl(\mp r\,\mathbf{n}\cdot\sigma\bigr) &= \sum_{\text{even}\,k}\frac{(\mp r)^k}{k!}\,(\mathbf{n}\cdot\boldsymbol{\sigma})^k\;+\;\sum_{\text{odd}\,k}\frac{(\mp r)^k}{k!}\,(\mathbf{n}\cdot\boldsymbol{\sigma})^k\\
&= \sum_{\text{even}\,k}\frac{(\mp r)^k}{k!}\times 1\;+\;\sum_{\text{odd}\,k}\frac{(\mp r)^k}{k!}\times(\mathbf{n}\cdot\boldsymbol{\sigma})\\
&= \cosh(r)\times 1\;\mp\;\sinh(r)\times(\mathbf{n}\cdot\boldsymbol{\sigma}).
\end{aligned}\tag{S.31}$$

In light of eqs. (9) relating the rapidity r to the β and γ parameters of the Lorentz boost, eq. (S.31) amounts to

$$\exp\bigl(\mp r\,\mathbf{n}\cdot\sigma\bigr) \;=\; \gamma\;\mp\;\beta\gamma(\mathbf{n}\cdot\boldsymbol{\sigma}) \;=\; \gamma\bigl(1\mp\vec{\beta}\cdot\boldsymbol{\sigma}\bigr).\tag{S.32}$$

Comparing this formula to the $\mathbf{2}$ and $\overline{\mathbf{2}}$ representations of the Lorentz boost, we arrive at

$$\bigl(M_L(B)\bigr)^2 \;=\; \gamma\bigl(1-\vec{\beta}\cdot\boldsymbol{\sigma}\bigr)\quad\text{and}\quad\bigl(M_R(B)\bigr)^2 \;=\; \gamma\bigl(1+\vec{\beta}\cdot\boldsymbol{\sigma}\bigr).\tag{S.33}$$

Finally, taking square roots of both sides of these formulae, we arrive at

$$M_L(B) \;=\; \sqrt{\gamma}\sqrt{1-\vec{\beta}\cdot\boldsymbol{\sigma}}\quad\text{and}\quad M_R(B) \;=\; \sqrt{\gamma}\sqrt{1+\vec{\beta}\cdot\boldsymbol{\sigma}}\,,\tag{10}$$

quod erat demonstrandum.

Problem **3**(d):

In the previous homework#6 (problem 4(b)), we saw that for any continuous Lorentz transform L,

$$M_R(L) \;=\; \sigma_2 M_L^*(L)\sigma_2\quad\text{and}\quad M_L(L) \;=\; \sigma_2 M_R^*\sigma_2\,.\tag{S.34}$$

Consequently, considering the transformation laws (11) of the Weyl spinors $\Psi_L(x)$ and $\Psi_R(x)$ and their complex conjugates, we see that

$$\begin{aligned}
\psi'_L(x') &= M_L\times\psi_L(x),\\
\psi'^*_L(x') &= M_L^*\times\psi_L(x),\\
\sigma_2\times\psi'^*_L(x') &= \sigma_2\times M_L^*\times\psi_L^*(x) \;=\; \sigma_2 M_L^*\sigma_2\times\sigma_2\psi_L^*(x) \;=\; M_R\times\sigma_2\psi_L^*(x),
\end{aligned}\tag{S.35}$$

thus $\sigma_2\times\psi_L^*(x)$ transforms under the continuous Lorentz transforms exactly like the $\psi_R(x)$.

Likewise,

$$\begin{aligned}
\psi'_R(x') &= M_R \times \psi_R(x), \\
\psi'^*_R(x') &= M^*_R \times \psi_R(x), \\
\sigma_2 \times \psi'^*_R(x') &= \sigma_2 \times M^*_R \times \psi^*_R(x) \;=\; \sigma_2 M^*_R \sigma_2 \times \sigma_2 \psi^*_R(x) \;=\; M_L \times \sigma_2 \psi^*_R(x),
\end{aligned} \tag{S.36}$$

thus $\sigma_2 \times \psi^*_R(x)$ transforms under the continuous Lorentz transforms exactly like the $\psi_L(x)$.

Problem **3**(e):

Given the γ^μ matrices (3) and the decomposition (11) of the Dirac spinor field $\Psi(x)$ into 2 Weyl spinor fields $\psi_L(x)$ and $\psi_R(x)$, we have

$$(i\gamma^\mu \partial_\mu \,-\, m)\Psi = \begin{pmatrix} -m & i\sigma^\mu \partial_\mu \\ i\overline{\sigma}^\mu \partial_\mu & -m \end{pmatrix} \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} \;=\; \begin{pmatrix} -m\psi_L \,+\, i\sigma^\mu \partial_\mu \psi_R \\ i\overline{\sigma}^\mu \partial_\mu \psi_L \,-\, m\psi_R \end{pmatrix} \tag{S.37}$$

while

$$\overline{\Psi} \;=\; \Psi^\dagger \gamma^0 \;=\; (\,\psi^\dagger_L \quad \psi^\dagger_R\,) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \;=\; (\,\psi^\dagger_R \quad \psi^\dagger_L\,)\,. \tag{S.38}$$

Consequently, the Dirac Lagrangian becomes

$$\begin{aligned}
\mathcal{L} &= \overline{\Psi}\,(i\gamma^\mu \partial_\mu \,-\, m)\Psi \\
&= \psi^\dagger_R\Big(-m\psi_L \,+\, i\sigma^\mu \partial_\mu \psi_R\Big) \;+\; \psi^\dagger_L\Big(i\overline{\sigma}^\mu \partial_\mu \psi_L \,-\, m\psi_R\Big) \\
&= i\psi^\dagger_R \sigma^\mu \partial_\mu \psi_R \;+\; i\psi^\dagger_L \overline{\sigma}^\mu \partial_\mu \psi_L \;-\; m\,\psi^\dagger_L \psi_R \;-\; m\,\psi^\dagger_R \psi_L\,.
\end{aligned} \tag{S.39}$$

Problem **3**(f):

For the massless fermions — and only for the massless fermions, — the last two terms in the Lagrangian (S.39) go away, and we are left with

$$\mathcal{L}_{\text{massless}} \;=\; i\psi^\dagger_L \overline{\sigma}^\mu \partial_\mu \psi_L \;+\; i\psi^\dagger_R \sigma^\mu \partial_\mu \psi_R \tag{S.40}$$

where the two Weyl spinor fields $\psi_L(x)$ and $\psi_R(x)$ are completely independent from each other. In particular, they obey independent Weyl equations

$$i\bar{\sigma}^\mu \partial_\mu \psi_L \;=\; 0 \quad \text{and} \quad i\sigma^\mu \partial_\mu \psi_R \;=\; 0. \tag{S.41}$$

On the other hand, for $m \neq 0$ the two last terms in the Lagrangian (S.39) connect the ψ_L and ψ_R to each other and we cannot have one without the other. In particular, their equations of motion become mixed,

$$i\bar{\sigma}^\mu \partial_\mu \psi_L \;=\; m\psi_R \quad \text{and} \quad i\sigma^\mu \partial_\mu \psi_R \;=\; m\psi_L\,. \tag{S.42}$$

<u>Problem **4**(a)</u>:

As we saw in class, the Dirac equation $(i\,\partial\!\!\!/ - m)\Psi = 0$ implies the Klein–Gordon equation $(\partial^2 + m^2)\Psi = 0$, hence any plane-wave solution of the Dirac equation must have on-shell momentum p^μ with $p^2 = m^2$. But there is more to the Dirac equation than the Klein–Gordon equation, hence eqs. (13) for the spinor coefficients of the $e^{\mp ipx}$ plane waves. Indeed, for $\Psi_\alpha(x) = e^{-ipx} u_\alpha$ the Dirac equation becomes

$$(i\,\partial\!\!\!/ - m) e^{-ipx} u \;=\; e^{-ipx} \times (p\!\!\!/ - m) u, \tag{S.43}$$

so the constant spinor u_α must obey the matrix equation $(p\!\!\!/ - m)u = 0$. Likewise, for $\Psi_\alpha(x) = e^{+ipx} v_\alpha$ the Dirac equation becomes

$$(i\,\partial\!\!\!/ - m) e^{+ipx} v \;=\; e^{+ipx} \times (-\,p\!\!\!/ - m) v, \tag{S.44}$$

so the spinor v_α must obey the matrix equation $(p\!\!\!/ + m)v = 0$. Conversely, for any on-shell momentum p and any spinors u_α and v_α obeying the matrix equations (13) the plane waves $e^{-ipx} u_\alpha$ and $e^{+ipx} v_\alpha$ obey the Dirac equation.

Problem $\mathbf{4}$(b):

For $\mathbf{p} = \mathbf{0}$, $p^0 = +m$ and $\not{p} - m = m(\gamma^0 - 1)$. Hence, the $u(\mathbf{p} = \mathbf{0}, s)$ spinors satisfy $(\gamma^0 - 1)u = 0$, or in the Weyl basis

$$\begin{pmatrix} -\mathbf{1}_{2\times 2} & \mathbf{1}_{2\times 2} \\ \mathbf{1}_{2\times 2} & -\mathbf{1}_{2\times 2} \end{pmatrix} u = 0 \quad \Longrightarrow \quad u = \begin{pmatrix} \zeta \\ \zeta \end{pmatrix} \tag{S.45}$$

where ζ is an arbitrary two-component spinor. Its normalization follows from $u^\dagger u = 2\,\zeta^\dagger\zeta$, so if we want $u^\dagger u = 2E_{\mathbf{p}} = 2m$ (for $\mathbf{p} = 0$), then we need $\zeta^\dagger\zeta = m$. Equivalently, we want $\zeta = \sqrt{m}\,\xi$ — and hence u as in eq. (15) — for a conventionally normalized spinor ξ with $\xi^\dagger\xi = 0$.

Note that there are two independent choices of ξ, normalized to $\xi_s^\dagger\xi_{s'} = \delta_{s,s'}$, so they give rise to two independent $u_\alpha(\mathbf{0}, s)$ spinors normalized to $u^\dagger(\mathbf{0}, s)u(\mathbf{0}, s') = 2m\delta_{s,s'} = 2E_{\mathbf{p}}\delta_{s,s'}$. They correspond to the two spin states of the $\mathbf{p} = 0$ electron. In terms of the spin vector, $\mathbf{S} = \frac{1}{2}\xi_s^\dagger\boldsymbol{\sigma}\xi_s$.

Problem $\mathbf{4}$(c):

The Dirac equation is Lorentz-covariant, so we may obtain solutions for all $p^\mu = (+E_{\mathbf{p}}, \mathbf{p})$ by simply Lorentz-boosting the solutions (15) from the rest frame where $p_0^\mu = (+m, \mathbf{0})$. Thus,

$$u(p, s) = M_D(B)\,u(p_0, s) = \begin{pmatrix} M_L & 0 \\ 0 & M_R \end{pmatrix} \begin{pmatrix} \sqrt{m}\,\xi_s \\ \sqrt{m}\,\xi_s \end{pmatrix} = \begin{pmatrix} \sqrt{m}\,M_L\,\xi_s \\ \sqrt{m}\,M_R\,\xi_s \end{pmatrix} \tag{S.46}$$

where M_D, M_L, and M_R are respectively Dirac-spinor, LH–Weyl-spinor, and RH–Weyl-spinor representations or the Lorentz boost B from $p^\mu = (m, \mathbf{0})$ to $p^\mu = (E, \mathbf{p})$. As we saw in problem 3(c), for a boost of velocity β in the direction $\mathbf{n}$,

$$M_L = \sqrt{\gamma} \times \sqrt{1 - \beta\mathbf{n}\cdot\boldsymbol{\sigma}}, \quad M_R = \sqrt{\gamma} \times \sqrt{1 + \beta\mathbf{n}\cdot\boldsymbol{\sigma}}\,. \tag{10}$$

For the boost in question

$$\gamma = \frac{E}{m}, \quad \gamma\beta\mathbf{n} = \frac{\mathbf{p}}{m}, \tag{S.47}$$

hence

$$\sqrt{m}\,M_L \;=\; \sqrt{E\,-\,\mathbf{p}\cdot\boldsymbol{\sigma}}\,, \quad \sqrt{m}\,M_R \;=\; \sqrt{E\,+\,\mathbf{p}\cdot\boldsymbol{\sigma}}\,. \tag{S.48}$$

Plugging these formulae into eq. (S.46) immediately gives us

$$u(p,s) \;=\; \begin{pmatrix} \sqrt{E-\mathbf{p}\cdot\boldsymbol{\sigma}}\,\xi_s \\ \sqrt{E+\mathbf{p}\cdot\boldsymbol{\sigma}}\,\xi_s \end{pmatrix}. \tag{16}$$

<u>Problem **4**(d)</u>:

The negative-frequency solutions $e^{+ipx}v_\alpha(p,s)$ have Dirac spinors v_α satisfying $(\not{p}+m)v=0$. For a particle at rest, $p^\mu=(+m,\mathbf{0})$, this equation becomes $m(\gamma^0+1)v=0$, or in the Weyl basis

$$\begin{pmatrix} \mathbf{1}_{2\times 2} & \mathbf{1}_{2\times 2} \\ \mathbf{1}_{2\times 2} & \mathbf{1}_{2\times 2} \end{pmatrix} v(\mathbf{p}=\mathbf{0},s) \;=\; 0 \quad\Longrightarrow\quad v(\mathbf{p}=\mathbf{0},s) \;=\; \sqrt{m}\begin{pmatrix} +\eta_s \\ -\eta_s \end{pmatrix} \tag{S.49}$$

for some two-component spinor η_s. As in part (b), the $\sqrt{m}$ factor translates the normalization $v^\dagger(p,s)v(p,s')=2E_p\delta_{s,s'}=2m\delta_{s,s'}$ (for $\mathbf{p}=\mathbf{0}$) to $\eta_s^\dagger\eta_{s'}=\delta_{s,s'}$.

For $\mathbf{p}\neq\mathbf{0}$ we proceed similarly to part (c), namely Lorentz-boost the rest-frame solution (S.49) to the frame where $p^\mu=(+E_\mathbf{p},\mathbf{p})$:

$$\begin{aligned} v(p,s) \;&=\; M_D(B)\,v(p_0,s) \;=\; \begin{pmatrix} M_L & 0 \\ 0 & M_R \end{pmatrix}\begin{pmatrix} +\sqrt{m}\,\eta_s \\ -\sqrt{m}\,\eta_s \end{pmatrix} \\ &=\; \begin{pmatrix} +\sqrt{m}\,M_L\,\eta_s \\ -\sqrt{m}\,M_R\,\eta_s \end{pmatrix} \;=\; \begin{pmatrix} +\sqrt{E-\mathbf{p}\cdot\boldsymbol{\sigma}}\,\eta_s \\ -\sqrt{E+\mathbf{p}\cdot\boldsymbol{\sigma}}\,\eta_s \end{pmatrix} \end{aligned} \tag{S.50}$$

in accordance with eq. (17)

Problem **4**(e):

Physically, a hole in the Fermi sea has opposite free energy, opposite momentum, opposite spin, *etc.*, from the missing fermion (see my notes for the explanation). A positron is a hole in the Dirac sea of electrons, so it should have opposite $p^\mu = (E, \mathbf{p})$ from the missing electron state — that's why the $v(p,s)$ spinors accompany the $e^{+ipx} = e^{+iEt-i\mathbf{px}}$ plane waves instead of the $e^{-ipx} = E^{-iEt+i\mathbf{px}}$ factors of the $u(p,s)$ spinors. Likewise, the positron should have the opposite spin state from the missing electron, and that's why the η_s should have the opposite spin from the ξ_s. More accurately, the η_s should carry the opposite spin vector $\eta_s^\dagger \mathbf{S} \eta_s$ from the $\xi_s^\dagger \mathbf{S} \xi_s$.

The solution to this spin relation is $\eta_s = \sigma_2 \xi_s^*$ (where $*$ denotes complex conjugation). To see how this works, note that σ_1 and σ_3 Pauli matrices are real while σ_2 is imaginary. Consequently,

$$\begin{aligned} \sigma_2 \sigma_2^* \sigma_2 &= -\sigma_2 \sigma_2 \sigma_2 = -\sigma_2 \,, \\ \sigma_2 \sigma_{1,3}^* \sigma_2 &= +\sigma_2 \sigma_{1,3} \sigma_2 = -\sigma_{1,3} \,, \end{aligned} \tag{S.51}$$

or in vector notations

$$\sigma_2 \boldsymbol{\sigma}^* \sigma_2 = -\boldsymbol{\sigma}. \tag{S.52}$$

Therefore, for

$$\eta = \sigma_2 \xi^* \quad \Longrightarrow \quad \eta^\dagger = \xi^\top \sigma_2 \,, \tag{S.53}$$

we have

$$\eta^\dagger \boldsymbol{\sigma} \eta = \xi^\top \sigma_2 \boldsymbol{\sigma} \sigma_2 \xi^* = \left(\xi^\dagger \sigma_2 \boldsymbol{\sigma}^* \sigma_2 \xi \right)^* = \left(-\xi^\dagger \boldsymbol{\sigma} \xi \right)^* = -\xi^\dagger \boldsymbol{\sigma} \xi \tag{S.54}$$

or in other words

$$\eta^\dagger \mathbf{S} \eta = \tfrac{1}{2} \eta^\dagger \boldsymbol{\sigma} \eta = -\xi^\dagger \mathbf{S} \xi \,. \tag{S.55}$$

And this is why we set $\eta_s = \sigma_2 \xi_s^*$.

Now consider implication of this relation between the ξ_s and η_s spinors for the plane-wave factors $u_\alpha(p,s)$ and $v_\alpha(p,s)$. Thanks to eq. (S.52) we have

$$\sigma_2 \times \sqrt{E \mp \mathbf{p}\cdot\boldsymbol{\sigma}} \times \sigma_2 \;=\; \left(\sqrt{E \pm \mathbf{p}\cdot\boldsymbol{\sigma}}\right)^*, \tag{S.56}$$

hence in light of the explicit formulae (16) and (17),

$$\begin{aligned} v(p,s) \;&=\; \begin{pmatrix} +\sqrt{E-\mathbf{p}\cdot\boldsymbol{\sigma}}\,\eta_s \\ -\sqrt{E+\mathbf{p}\cdot\boldsymbol{\sigma}}\,\eta_s \end{pmatrix} \;=\; \begin{pmatrix} +\sqrt{E-\mathbf{p}\cdot\boldsymbol{\sigma}}\,\sigma_2\xi_s^* \\ -\sqrt{E+\mathbf{p}\cdot\boldsymbol{\sigma}}\,\sigma_2\xi_s^* \end{pmatrix} \\ &=\; \begin{pmatrix} +\sigma_2\sigma_2\sqrt{E-\mathbf{p}\cdot\boldsymbol{\sigma}}\,\sigma_2\xi_s^* \\ -\sigma_2\sigma_2\sqrt{E+\mathbf{p}\cdot\boldsymbol{\sigma}}\,\sigma_2\xi_s^* \end{pmatrix} \;=\; \begin{pmatrix} +\sigma_2\left(\sqrt{E+\mathbf{p}\cdot\boldsymbol{\sigma}}\right)^*\,\xi_s^* \\ -\sigma_2\left(\sqrt{E-\mathbf{p}\cdot\boldsymbol{\sigma}}\right)^*\,\xi_s^* \end{pmatrix} \\ &=\; \begin{pmatrix} 0 & +\sigma_2 \\ -\sigma_2 & 0 \end{pmatrix} \times \begin{pmatrix} +\sqrt{E-\mathbf{p}\cdot\boldsymbol{\sigma}}\,\xi_s \\ +\sqrt{E+\mathbf{p}\cdot\boldsymbol{\sigma}}\,\xi_s \end{pmatrix}^* \\ &=\; \gamma^2 u^*(p,s). \end{aligned} \tag{S.57}$$

This verifies the first eq. (18). We may verify the second eq. (18) in a similar manner, but it's easier to use γ^2 being imaginary and squaring to -1, hence $\gamma^2(\gamma^2)^* = -\gamma^2\gamma^2 = +1$, and therefore

$$\gamma^2 v^*(p,s) \;=\; \gamma^2\left(\gamma^2 u^*(p,s)\right)^* \;=\; \gamma^2\left(\gamma^2\right)^* u(p,s) \;=\; +u(p,s). \tag{S.58}$$

<u>Problem **4**(f)</u>:
The 3D spinors ξ_λ of definite helicity $\lambda = \mp\frac{1}{2}$ satisfy

$$(\mathbf{p}\cdot\boldsymbol{\sigma})\xi_\mp \;=\; \mp|\mathbf{p}| \times \xi_\mp\,. \tag{S.59}$$

Plugging these ξ_λ into the positive-energy Dirac spinors (16), we obtain

$$u(p,\lambda=\mp\tfrac{1}{2}) \;=\; \begin{pmatrix} \sqrt{E\pm|\mathbf{p}|}\times\xi_\mp \\ \sqrt{E\mp|\mathbf{p}|}\times\xi_\mp \end{pmatrix}. \tag{S.60}$$

In the ultra-relativistic limit $E\approx|\mathbf{p}|\gg m$, the square roots here simplify to $\sqrt{E+|\mathbf{p}|}\approx\sqrt{2E}$ and $\sqrt{E-|\mathbf{p}|}\approx 0$ (in comparison with the other root). Consequently, eq. (S.60) simplifies to

$$u(p,L) \;\approx\; \sqrt{2E}\begin{pmatrix}\xi_L\\ 0\end{pmatrix}, \qquad u(p,R) \;\approx\; \sqrt{2E}\begin{pmatrix}0\\ \xi_R\end{pmatrix}. \tag{S.61}$$

In other words, the ultra-relativistic positive-energy Dirac spinors of definite helicity are chiral — dominated by the LH Weyl components for the left helicity or by the RH Weyl components for the right helicity.

Now consider the negative-energy Dirac spinors (17). Because the η_s spinors have exactly opposite spins from the ξ_s, their helicities are also opposite and hence

$$(\mathbf{p}\cdot\boldsymbol{\sigma})\eta_{\mp} \;=\; \pm|\mathbf{p}|\times\eta_{\mp} \tag{S.62}$$

— note the opposite sign from eq. (S.59). Therefore, the negative-energy Dirac spinors v of definite helicity are

$$v(p,\lambda=\mp\tfrac{1}{2}) \;=\; \begin{pmatrix}+\sqrt{E\mp|\mathbf{p}|}\times\eta_{\mp}\\ -\sqrt{E\pm|\mathbf{p}|}\times\eta_{\mp}\end{pmatrix}, \tag{S.63}$$

and in the ultra-relativistic limit they become

$$v(p,L) \;\approx\; -\sqrt{2E}\begin{pmatrix}0\\ \eta_L\end{pmatrix}, \qquad v(p,R) \;\approx\; +\sqrt{2E}\begin{pmatrix}\eta_R\\ 0\end{pmatrix}. \tag{S.64}$$

Again, the ultra-relativistic negative-energy spinors are chiral, but this time the chirality is opposite from the helicity — the left-helicity spinor has dominant RH Weyl components while the right-helicity spinor has dominant LH Weyl components.